Studio North: is growth becoming more valuable?

Customer acquisition, monthly renewal and contribution economics | 24 monthly acquisition cohorts, 2024-01 through 2025-12; data cutoff 2026-01-01

THE BUSINESS AND THE DECISION

Studio North sells a monthly creative-work membership to independent professionals. Standard and premium plans combine ready-to-use templates with guided learning; members can also purchase workshops and other add-ons. Each month, members choose whether to renew. Search campaigns and partner workshops bring customers with two stated priorities: save time or build skills.

Management needs to know whether successive acquisition cohorts create more durable contribution, at a cost the company can recover quickly enough to fund growth. Company scale, customer quality and marketing incrementality answer different parts of that decision.

The matched-age endpoints show lower blended CAC, higher contribution and stronger cash recovery. Renewal is not clearly improving across successive cohorts; short campaign tests do not yet establish a profitable acquisition expansion.

ACQUISITION EFFICIENCY
$106 → $73
blended CAC per acquired customer

CUSTOMER VALUE
$123 → $144
six-month contribution per customer

CASH RECOVERY
$18 → $71
six-month cash after blended CAC

Matched-age comparison: 2024-01 to 2025-07. Values above are rounded to dollars; the evidence table reports cents.

WHAT THE EVIDENCE ESTABLISHES

Evidence	Earlier	Later	Interpretation
Sixth-month retained share	63.1%	67.9%	Observed at age index 5
Blended CAC	$105.79	$72.99	Tracked spending / all acquired people
Six-month contribution / customer	$123.39	$144.05	Cumulative, after variable costs
Six-month cash after acquisition	$17.60	$71.06	Per original acquired customer
Joining cohort	2024-01	2025-07	Latest cohort with six observed ages

RECOMMENDED MANAGEMENT RESPONSE

Action. Marketing and Analytics should preserve controls and extend both campaign follow-ups through the cash-recovery horizon. Finance should monitor six-month cash against the trailing six mature cohorts' median of $50.36 per customer, not use it as an automatic causal gate. Product should investigate the Search / save-time renewal weakness before assuming conversion gains imply durable value. Incremental contribution after media per assigned prospect: search / save time: $1.16 over 31 days (95% interval $-0.62 to $2.94); partner / build skill: $-0.36 over 30 days (95% interval $-2.26 to $1.54). 2 of 2 intervals include zero; neither the sign nor the ranking of point estimates warrants a scale-or-stop decision.

WHAT WOULD CHANGE THE DECISION

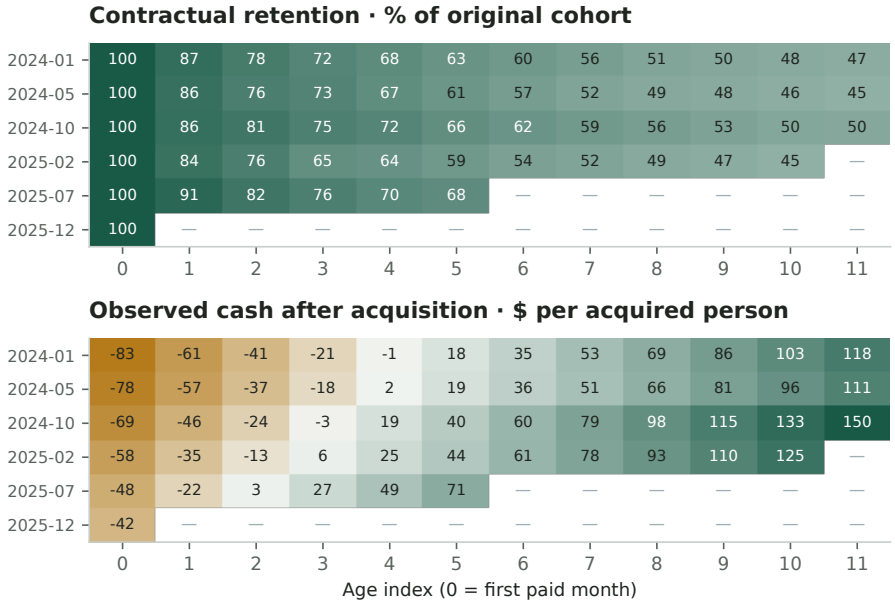
For immature cohorts, value depends on future renewals and spending. Treat their LTV as a conditional forecast. Scale a campaign only when a randomized comparison supports incremental contribution over a relevant window; historical attributed CAC alone cannot establish that return. Retain the cost and cash-recovery constraints when a campaign produces more revenue.

All customer records describe an illustrative business and are synthetic. The analysis covers recorded variable costs and acquisition spending, not fixed operating costs, financing or taxes. It therefore does not establish company profitability.

The management pages summarize the diagnosis and action. Appendices document the model, parameter estimates, validation, data architecture and dataset-level Universal Customer View.

Compare customer quality at a common age

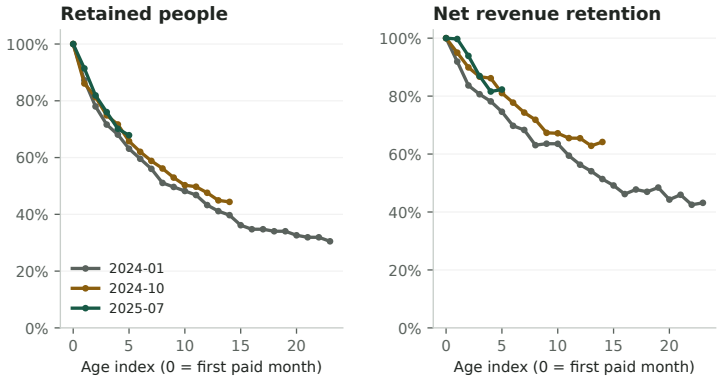
A joining cohort contains customers whose first successful payment occurred in the same month. Future ages remain unobserved.



Saved SQL outputs: `cohort_totals.csv` and `cohort_cash_economics.csv`. Age 0 is the first paid month. Cash balance is cumulative contribution less tracked acquisition spending, divided by original cohort size. These six cohorts are an evenly spaced subset of the sorted 24-cohort sequence, including its endpoints; they were not selected for favorable results.

ACQUISITION EFFICIENCY AND DURABILITY MUST MOVE TOGETHER

At the common sixth age month, retained share is higher: 63.1% for 2024-01 versus 67.9% for 2025-07. Blended CAC is lower: $105.79 versus $72.99. It divides tracked spending by all acquired people; organic control buyers enter that denominator in cohorts where they occur. Paid-source CAC has a different denominator. These are descriptive comparisons, not campaign effects. At the same age, cumulative contribution per acquired customer changes from $123.39 to $144.05; cash after acquisition changes from $17.60 to $71.06. Both calculations use recorded variable costs.

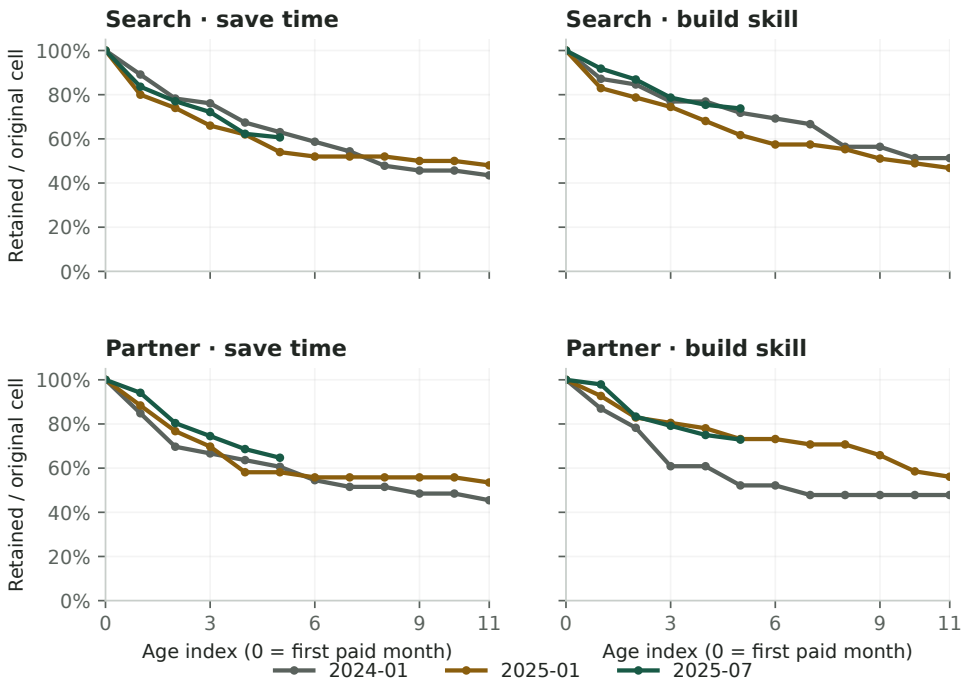


Retained share uses original cohort membership; revenue retention uses each cohort's first-period revenue. Revenue can change because of the number of renewing members, their plan mix or add-on spending. Product activity is not a substitute for a renewal record.

The trajectory matters as well as the cutoff. A stronger first month does not establish longer customer life, while a customer-count decline can coexist with more spending by the members who remain.

Check whether progress survives the customer mix

Observed renewal trajectories by acquisition channel and stated need; each panel uses the same axes.



Saved SQL output: `cohort_age_summary.csv`. Curves divide retained members by the original cohort–channel–need cell. No missing future observations are filled with zeros.

WITHIN-GROUP DIAGNOSIS

The two displayed endpoints improve in 3 of 4 channel-by-need cells, but the intervening curves do not establish a steady renewal trend. Search / save time changes from 63% to 61% retained at age 5. Its short-window experimental conversion response does not establish better subsequent renewal.

A USEFUL SEGMENTATION SEPARATES RELEVANT DIFFERENCES

The primary need grouping has the lowest within-group variance in six-month contribution (4,414.7 dollars squared), but the reduction versus the next-best scheme is 2.3%. This is a modest separation, with substantial residual customer heterogeneity. Use the groups to organize the diagnosis; a need label does not make their economics homogeneous.

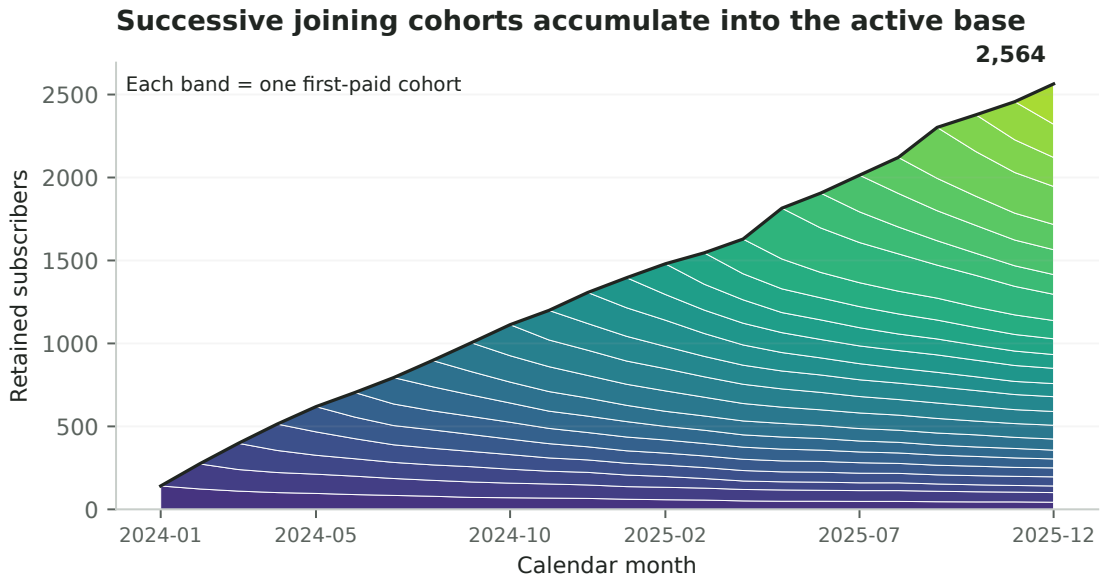
Grouping	Variance (dollars squared)	Groups	Smallest group
age band	4,520.2	2	1718
primary need	4,414.7	2	1578
region	4,520.3	2	1725

Comparison uses the same mature-customer sample and the same six-month outcome. Smaller within-group variance means more homogeneous observed outcomes within the chosen groups; it does not imply identical customers or establish a causal effect of a need or channel.

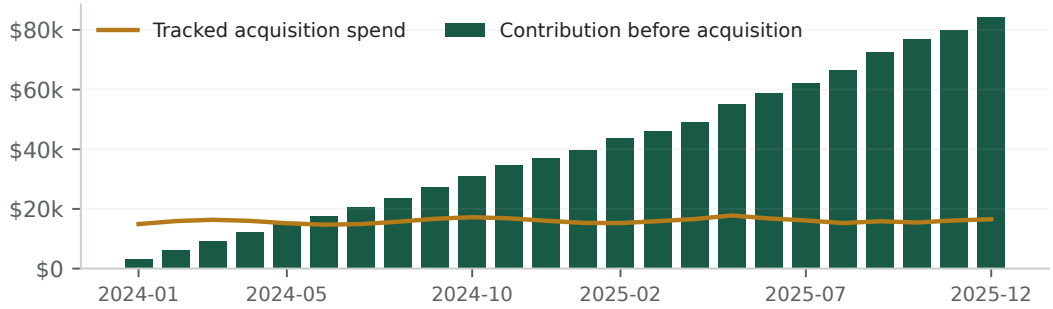
Stated need is known before acquisition. It can support targeting and journey interpretation. Joining month defines a cohort, not a segment. Survey evaluations and observed behavior can suggest explanations to test, but neither proves why a customer renewed or left.

Retained cohorts determine how growth accumulates

The company is the sum of its acquisition cohorts at different ages, not the latest cohort scaled up.



Saved SQL output: `company_month.csv`. Each layer is a joining cohort's active customers at the displayed calendar month. Layer totals are reconciled to independently saved company totals before plotting.



SCALE AND CASH RECOVERY ARE SEPARATE TESTS

New acquisition adds customers immediately. Renewals preserve older layers and their contribution, reducing the replacement acquisition required to sustain scale. The lower chart compares recorded contribution with acquisition spending in each calendar month; it is not cumulative retained customers or a profit statement. In 2025-12, recorded contribution was $84,335 and acquisition spending was $16,529, leaving $67,806 before fixed costs, financing and taxes. This period result should be read alongside the age-aligned cohort comparison, rather than used as a substitute for it.

OPERATING IMPLICATION

Track both the common-age economics of each joining cohort and the company's accumulated contribution. A favorable acquisition metric can coexist with slower cash recovery when renewals weaken. A growing customer base can also conceal deterioration if management replaces departing customers with increasingly costly new ones.

Cash payback is undiscounted: the first age at which cumulative contribution covers the acquisition investment. The LTV calculation that follows discounts expected monthly contribution and uses a common, explicit horizon.

Forecast monthly renewal before forecasting value

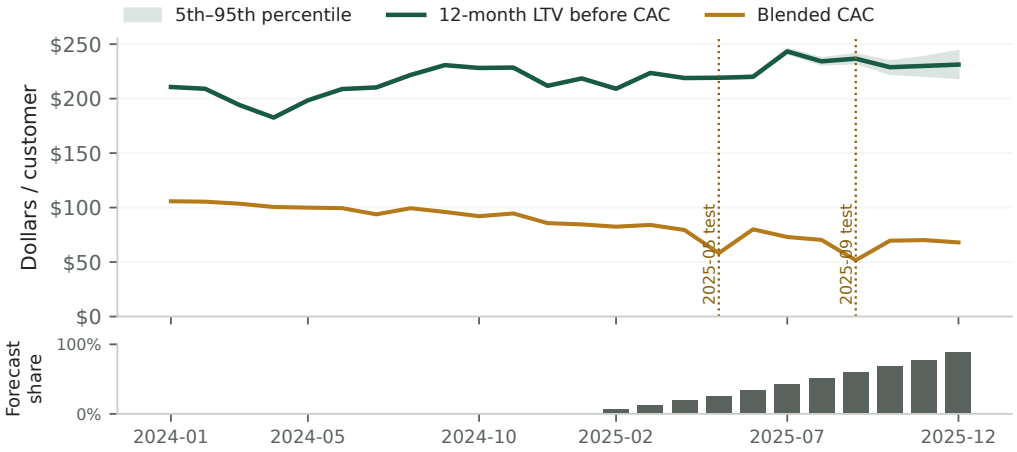
Hierarchical shifted-Beta-Geometric retention + Gamma gross revenue; NumPyro NUTS

HOW THE MODEL REPRESENTS THE BUSINESS

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$\text{net LTV}_H = \sum_{a=0}^{H-1} \frac{\text{E}[\text{contribution}_a \mid \text{observed history}]}{(1+d)^{a+1}} - \text{CAC}\,.$$

Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Keep observed revenue and contribution; forecast recursively from the last observed survivors. All cohorts are valued over the same 12 monthly receipts, discounting at 1% per month. Forecast bands and the probability of nonnegative expected cash cover parameter uncertainty, not renewal/revenue process noise or all future business risk. The main retention model extends Fader–Hardie's shifted-Beta-Geometric approach; the revenue model adapts Orduz's Gamma structure. An Orduz-inspired neural retention model is a separate validation benchmark, not the source of executive LTV. Parameter estimates and held-out checks appear in the appendix.



Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval. The lower strip shows the forecast share of common 12-month LTV. Dotted markers identify experiment months with organic control buyers in the denominator. Blended CAC divides tracked spending by all acquired customers; it is not the causal cost of the next customer.

MANAGEMENT INTERPRETATION

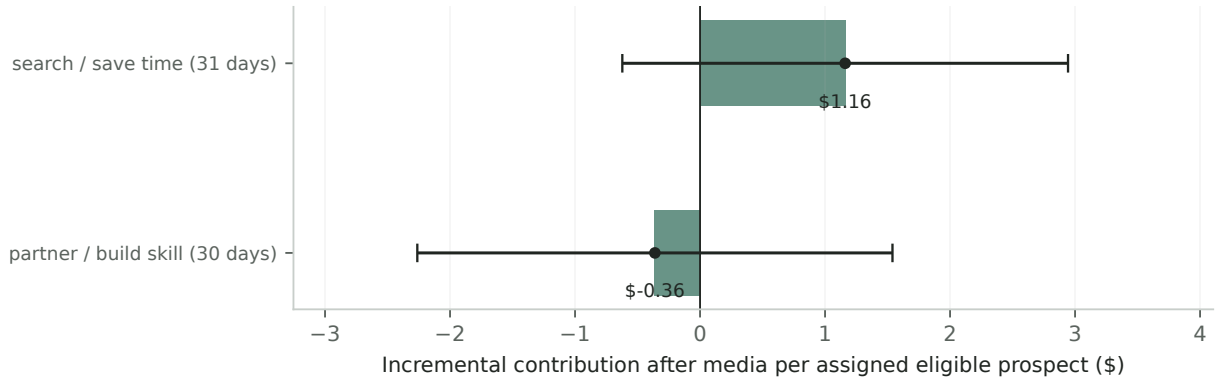
Across the cohort sequence, common-horizon contribution LTV combines observed receipts with a growing predicted remainder; the strip makes that dependence visible. Net of blended CAC, the endpoint comparison is $104.90 for 2024-01 versus $163.21 for 2025-12. This is not a monotonic improvement claim. The newest value is 89% forecast, with a 5th–95th parameter range of $150.96 to $175.83.

Future net revenue and contribution shares of payment revenue use observed channel-by-need-by-age shares, with the same group's observed renewal ages (age>=1) as fallback. Shares are fixed assumptions, not causally identified or assigned posterior uncertainty. Predicted customer value is not experimentally identified campaign return. A favorable average LTV-to-CAC comparison still requires a cash-recovery constraint and evidence that the next acquisition investment creates additional value.

Require the next campaign to beat its own baseline

Randomized assignment includes all eligible prospects, including those who never purchase. Each test keeps its own control and follow-up window.

Test	Arm	Assigned	Buyers	Contribution	Media
search / save time	control	250	30	$739.00	$0.00
search / save time	treatment	250	70	$1,679.25	$650.00
partner / build skill	control	250	34	$834.85	$0.00
partner / build skill	treatment	250	62	$1,644.55	$900.00



Saved SQL outputs: `experiment_summary.csv` and `experiment_effects.csv`. Whiskers are approximate 95% intervals from all-assigned contribution variances, including nonbuyer zeros. Fixed media costs shift the mean, not its sampling variance. These 30/31-day windows are shorter than the cohorts' cash-recovery horizon; they do not measure lifetime campaign return.

THE SAME ECONOMICS APPLY TO TREATMENT AND CONTROL

For each arm, net revenue less recorded variable costs gives contribution. Compare contribution per assigned eligible prospect, then subtract the treatment's additional media cost. Control purchases remain in the baseline rather than receiving campaign credit.

$$\Delta\text{contribution after media per prospect} = \frac{C_T - M_T}{N_T} - \frac{C_C - M_C}{N_C}.$$

MANAGEMENT ACTIONS AND ACCOUNTABILITY

Accountable team	Next action	Decision evidence
Marketing + Finance	Defer expansion or rejection based only on the short-window profit estimates; monitor recovery against mature-cohort experience.	Age-aligned value and cash, alongside uncertainty in incremental returns.
Product + Operations	Investigate Search / save-time renewal weakness despite its positive short-term conversion response.	Renewal events, product actions and expressed customer evaluations.
Marketing + Analytics	Preserve both controls and extend follow-up through the acquisition cash-recovery horizon.	Contribution after media per assigned prospect, including nonbuyers, with uncertainty.
Data + Analytics	Refresh the evidence from the source exports and review forecast misses before releasing the decision report.	SQL reconciliation, model holdouts, parameter output and file hashes.

Do not multiply experimental conversion lift by the predicted LTV of observed buyers and label the product an established lifetime campaign effect. The tested population, outcome window and valuation assumptions must remain visible.

Parameter meanings and fitted estimates

Technical appendix | The inference model is distinct from the process used to generate the synthetic customer records.

LIKELIHOOD, HETEROGENEITY AND ECONOMIC CONVERSION

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$p_{i,g} \sim \mathrm{Beta}(\alpha_g, \beta_g), \qquad \alpha_g = \mu_g \phi_g, \qquad \beta_g = (1-\mu_g)\phi_g,$$
$$N_{g,t} \mid N_{g,t-1} \sim \mathrm{Binomial}(N_{g,t-1}, 1-h_g(t)), \quad h_g(t) = \frac{\alpha_g}{\alpha_g+\beta_g+t-1}.$$

g indexes a cohort–need–channel cell; $t=1$ is its first renewal. Mean-churn coefficients use a logit link; concentration and revenue use log links. Search and save-time customers are the reference groups. Retention cohort trend is $(\text{index}-11.5)/11.5$; revenue cohort trend is $(\text{index}-11.5)/12$. Revenue log-age is $\log(1+\text{age})$. Cohort index runs from 0 to 23.

Parameter	Prior	Mean	5th–95th	R-hat	ESS
cohort_scale	HalfNormal(0.25)	0.055	0.006 to 0.115	1.001	1,412
concentration_beta[intercept]	Normal(log(4),1)	1.450	1.249 to 1.659	1.000	2,556
concentration_beta[build_skill]	Normal(0,0.5)	0.232	-0.038 to 0.505	1.001	3,170
concentration_beta[partner]	Normal(0,0.5)	0.042	-0.227 to 0.319	1.000	2,896
concentration_beta[organic]	Normal(0,0.5)	-0.087	-0.832 to 0.655	1.000	5,560
mean_beta[intercept]	Normal(-2.5,1)	-1.818	-1.916 to -1.718	1.000	2,369
mean_beta[build_skill]	Normal(0,0.5)	-0.307	-0.429 to -0.185	1.001	3,381
mean_beta[partner]	Normal(0,0.5)	-0.076	-0.195 to 0.043	1.000	2,791
mean_beta[organic]	Normal(0,0.5)	0.078	-0.309 to 0.452	1.000	5,229
mean_beta[cohort_trend]	Normal(0,0.5)	-0.025	-0.110 to 0.059	1.000	4,770
revenue_beta[intercept]	Normal(-0.5,0.8)	-0.601	-0.606 to -0.596	1.000	6,944
revenue_beta[log1p_age]	Normal(0.1,0.8)	0.106	0.104 to 0.109	0.999	5,974
revenue_beta[cohort_trend]	Normal(0,0.8)	0.021	0.017 to 0.026	0.999	7,944
revenue_beta[build_skill]	Normal(0,0.8)	0.056	0.052 to 0.060	1.000	8,898
revenue_beta[partner]	Normal(0,0.8)	-0.002	-0.006 to 0.003	0.999	7,312
revenue_beta[organic]	Normal(0,0.8)	-0.019	-0.043 to 0.005	0.999	6,587
revenue_shape	LogNormal(log(4),0.8)	17.309	16.192 to 18.469	0.999	7,994

The 24 standardized cohort offsets z have posterior means from -0.691 to 0.769; multiply each draw by cohort_scale to obtain its log-odds shift. Their largest R-hat is 1.001; smallest effective sample size is 4,250. The complete parameter_summary.csv provides every cohort effect, prior, interval and interpretation.

At the coefficient means, Search / save time at centered cohort trend and zero offset has mean latent monthly churn mu=14.0% and concentration phi=4.26. The cohort-trend coefficient's 5th–95th interval is -0.110 to 0.059; it includes zero and does not establish a renewal improvement trend.

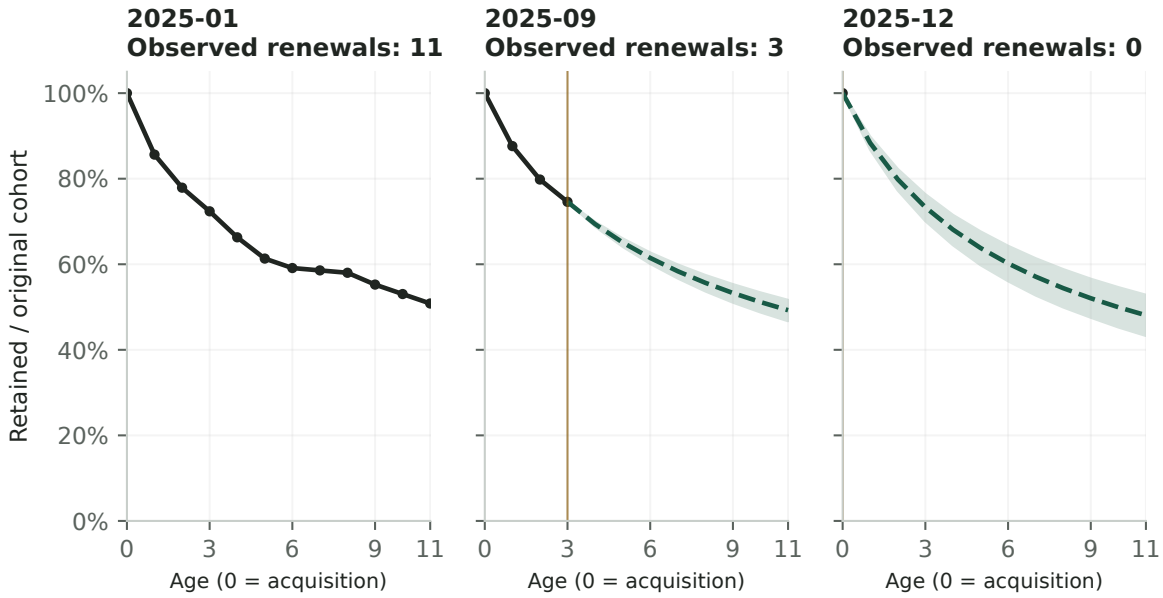
Posterior means and 5th–95th percentiles are read from the saved, hash-bound parameter output. Units and reference groups are stated with the parameter definitions; transformed parameters are not interchangeable with their latent log-scale values.

ESTIMATION AND REPRODUCIBILITY

Setting or diagnostic	Recorded value
Inference	NUTS
Chains / warmup / samples	4 / 1000 / 1000 per chain
Posterior forecast draws	400
Maximum R-hat / minimum ESS	1.001 / 1,412
Divergences	0
Seed / numerical backend	20261116 / cpu / float64

Show the unobserved renewal path explicitly

Technical appendix | The same model forecasts retained customers, then converts that path into revenue and contribution.



Saved model output: `forecast_curve.csv`. Each panel uses its original acquired-customer denominator. Solid points are observed; dashed lines are forecast expected retention; shading is the 5th–95th posterior parameter range. The vertical marker is the observation boundary.

THE AMOUNT OF OBSERVED RENEWAL HISTORY MATTERS

The complete cohort supplies an observed reference. The intermediate cohort has some renewal history but still requires extrapolation. The newest cohort has only its acquisition month: it has **zero observed renewal decisions**. Its future path borrows information from older cohorts, stated needs, acquisition channels and the model's partially pooled cohort distribution.



Observed contribution is retained; only the unobserved remainder is predicted. The newest cohort's forecast relies on older cohorts until its own renewal history becomes available.

These are expected customer paths conditional on fitted parameters. The bands exclude future customer-level Bernoulli variation, uncertain economic conversion and changes in the business regime. A narrow parameter band is not evidence that those risks are small.

Separate fitted agreement from prediction performance

Technical appendix | A model can describe observed counts closely and still miss the future.



Saved SQL/model output: `model_fit_checks.csv`. These are in-sample, one-step fitted residuals conditional on previous observed customer counts. They are not held-out forecast errors.

HELD-OUT PERFORMANCE

Training includes 2025-09; withheld months are 2025-10, 2025-11, 2025-12.

Withheld cells: 258. Retained-count mean absolute error: 1.11. Revenue weighted absolute percentage error: 6.1%. A separate training-only forecast checks complete 12-month values: LTV mean absolute error $3.13 versus $25.21 for its baseline. These tests evaluate prediction, not campaign effects.

Measure	Model	Baseline	Interpretation
Retained-count MAE	1.11	1.89	People per withheld cell
Retained-count WAPE	4.7%	Not reported	Retained customers, not revenue
Revenue WAPE	6.1%	10.8%	Withheld calendar months
12-month LTV MAE	$3.13	$25.21	Value per original customer

On the same 258 held-out cells, retained-count MAE is 1.112 for sBG, 1.146 for the Orduz-inspired 4-hidden-unit network, and 1.894 for the naive baseline. The two fitted models have similar retention accuracy in this holdout. The network uses variational inference and a conditional monthly churn likelihood; it does not supply executive LTV. Revenue WAPE below evaluates the main Gamma revenue model, not the retention-only network.

WHERE CALENDAR-TIME FIT DEPARTS MOST

Calendar month	Observed renewals	One-step fitted	Standardized residual
2025-04	1,420	1,442.2	-2.27
2025-03	1,357	1,378.0	-2.18
2024-08	739	725.6	+1.71
2025-08	1,904	1,885.8	+1.68

Four months with the largest absolute standardized calendar residual, selected from saved `calendar_fit_checks.csv`; only renewal ages 1 and older enter, not new acquisitions. These flag timing patterns, not causes, held-out errors or formal significance tests.

INTERVAL SCOPE AND REMAINING RISK

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval.

One historical year-end origin, not proof of accuracy for every future cohort or business regime.

Show exactly how much of each value was forecast

Technical appendix | Training includes the calendar month beginning 2024-12-01; later calendar months are used only for scoring.

At the 2024-12 forecast origin, 11 cohorts had an unfinished 12-month value horizon. Payback had already occurred for 7; those cases are excluded from forecast-month MAE. The 4 remaining cases with both observed and predicted recovery have payback-month MAE 0.00. This is a small future-payback sample, not evidence of universally exact prediction.

Cohort	Observed / forecast months	Actual LTV	Model LTV	Baseline LTV	Payback actual / model	Payback evaluation
2024-02	11 / 1	$209.05	$209.39	$209.24	6 / 6	Already observed
2024-03	10 / 2	$194.17	$193.97	$193.73	6 / 6	Already observed
2024-04	9 / 3	$182.60	$182.92	$179.44	6 / 6	Already observed
2024-05	8 / 4	$198.46	$197.02	$190.96	5 / 5	Already observed
2024-06	7 / 5	$208.79	$206.53	$198.83	5 / 5	Already observed
2024-07	6 / 6	$210.28	$212.54	$202.16	5 / 5	Already observed
2024-08	5 / 7	$221.72	$217.35	$198.93	5 / 5	Already observed
2024-09	4 / 8	$230.76	$231.85	$206.95	5 / 5	Future; scored
2024-10	3 / 9	$228.15	$219.39	$178.14	5 / 5	Future; scored
2024-11	2 / 10	$228.50	$225.81	$181.08	5 / 5	Future; scored
2024-12	1 / 11	$211.76	$222.48	$107.91	4 / 4	Future; scored

Saved model output: `valuation_backtest.csv`. LTV values are discounted contribution before blended CAC, per original acquired customer. “Observed / forecast” counts the receipts within the common 12-month horizon; payback month counts receipts from 1.

WHY TOTAL-VALUE AND FORECAST-TAIL ERRORS ARE THE SAME

Observed contribution is copied identically into the actual and predicted totals. The forecast covers only the remaining months. Consequently, total LTV error equals the error in the unobserved tail; the long observed prefix does not create additional predictive accuracy. Rows are ordered by the amount of history available at the origin.

THE BASELINE AND ITS WEAKEST-INFORMATION CASE

The naive baseline freezes the last three observed renewal rates and revenue per retained customer. A cohort with only its acquisition month has no observed renewal rate; the baseline uses an explicit 0.8 renewal fallback. Its value is an assumption, not an estimated renewal rate. The December row makes this low-information comparison visible.

WHAT THE PAYBACK FORECAST REPRESENTS

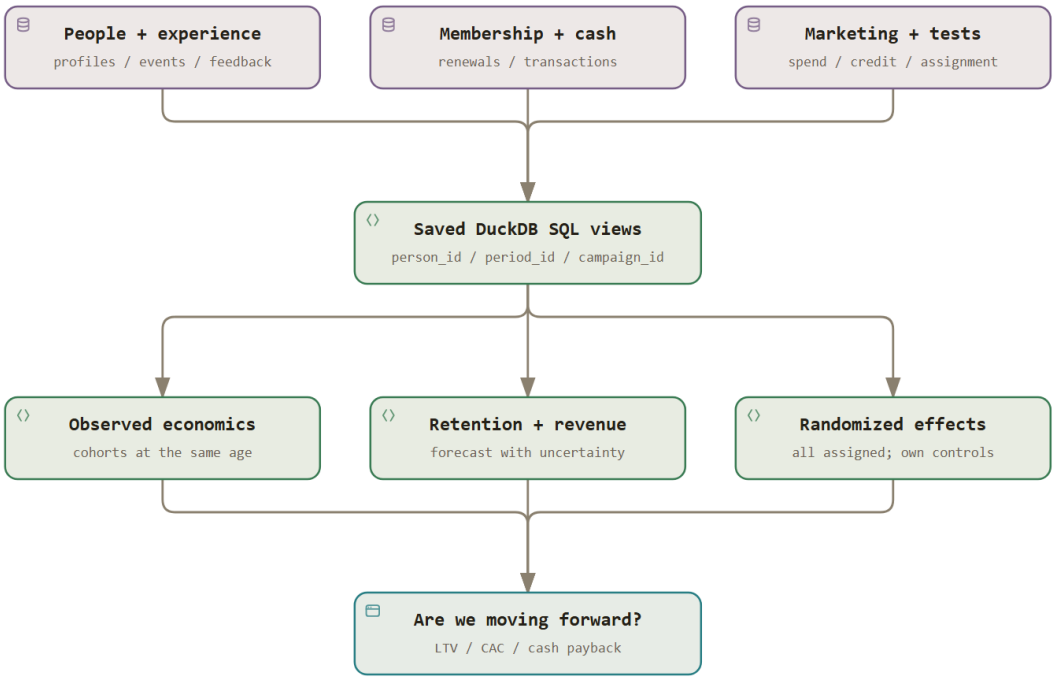
Forecast payback is the first crossing of acquisition cost by the **posterior-mean expected contribution path**. It is not the mean or median of a payback-time distribution, and it does not estimate the chance that a realized customer cohort recovers its cost by that month. Posterior parameter bands also exclude future customer-level process noise.

> Both the generator and sBG model contain persistent individual churn heterogeneity. The generator additionally uses a mixture of churn types, calendar shocks, cohort-specific dispersion, plan upgrades and add-on purchases. The inference model does not reproduce those mechanisms. The small set of future-payback cases is a narrow diagnostic under that deliberate mismatch: interpret zero month error, if observed, alongside the full value errors and incomplete histories.

Preserve source meaning through the joins

Technical appendix | Eight separate exports become tested analytical views in DuckDB; no database server is required.

Eight source exports; 24 monthly acquisition cohorts; cutoff 2026-01-01. Pending renewal outcomes are excluded from observed renewal trials. All randomized prospects remain in experimental denominators, including nonbuyers.



The diagram is rendered by the bundled Archify package from original source specifications. The accompanying `diagrams/source-tables.html` and SVG show the full table-level schema and cardinality. Dataset classifications on the following pages list concrete fields.

KEYS DO NOT MAKE UNLIKE RECORDS INTERCHANGEABLE

A person can have many renewal periods, ledger transactions, product events and survey responses. Those tables must be aggregated at the intended grain before they are joined. Acquisition spending is a campaign-level record; it does not become a customer-level observation merely because it can be allocated across acquired customers. Experimental assignment defines an eligible population that includes noncustomers.

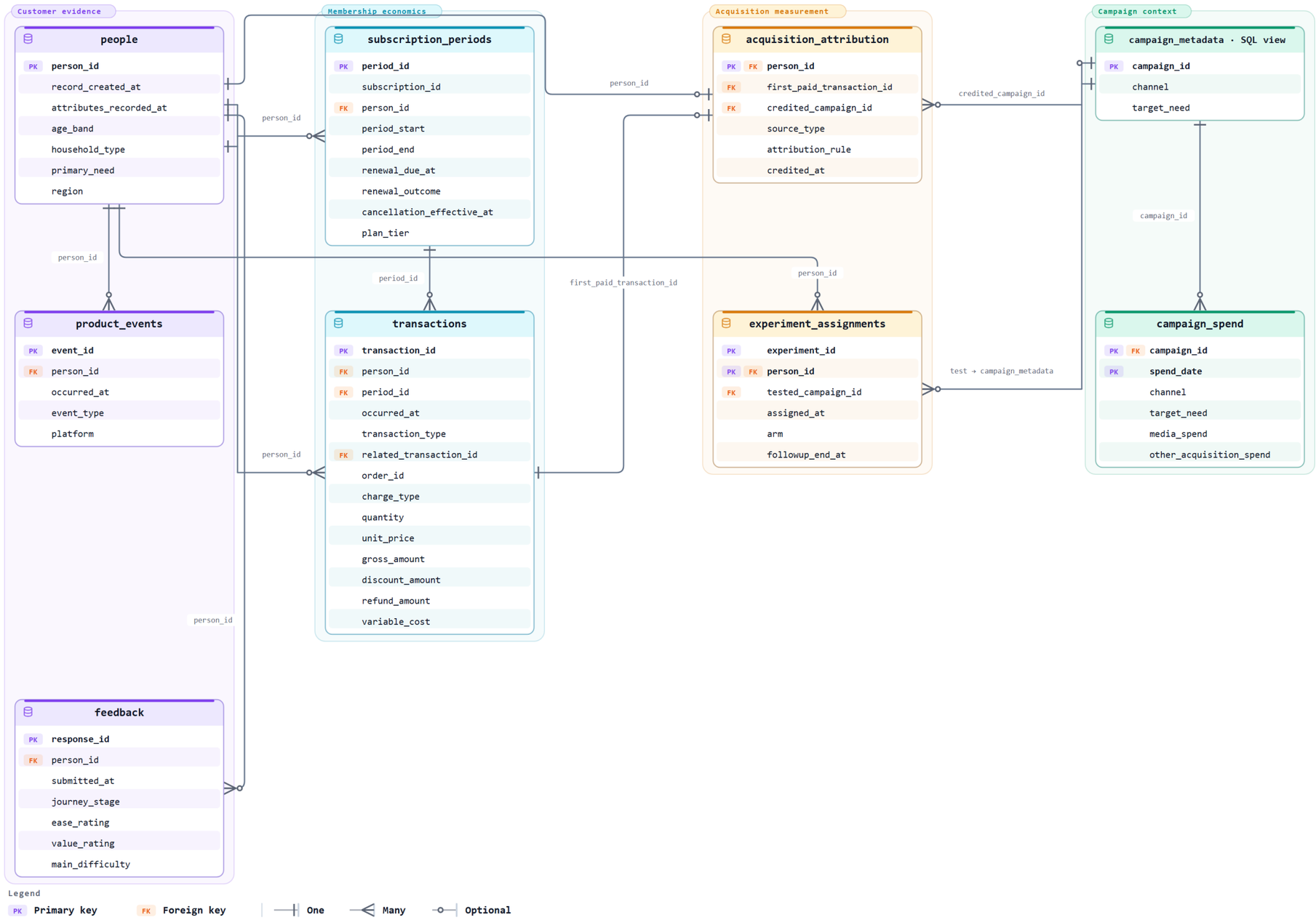
ANALYTICAL CONTRACTS

Definition	Contract
Joining cohort	Month of the first successful payment, derived from complete payment history.
Customer survival	Renewing contractual membership; original cohort size is the retained-share denominator.
Revenue and contribution	Ledger payments less discounts/refunds; contribution also subtracts recorded variable costs.
Acquisition investment	Tracked spend, including campaigns with no attributed acquisitions; no invented organic-source CAC.
Experimental unit	Every assigned eligible prospect, including nonbuyers and control buyers, within the prespecified follow-up window.
Unknown future	Right-censored ages stay unobserved. Missing survey responses remain missing perceptions.

Source hashes, saved SQL and downstream manifests bind the evidence chain. Numeric tables, editable LaTeX, source diagrams and the maintained style package accompany the report.

Source records, keys and cardinality

A3 landscape foldout | The eight raw exports remain separate. Campaign metadata is a derived lookup for a nonunique campaign-spend key.



Native Archify schema; editable SVG/HTML: diagrams/source-tables. Refunds link transactions.related_transaction_id to transactions.transaction_id within the same ledger. Campaign foreign keys target unique campaign_metadata; campaign_spend uses campaign_id + spend_date. Aggregate to the intended grain before joining one-to-many records.

Classify each source by what it actually measures

Technical appendix | Identifiers include keys and profile descriptors; engagement records behavior; relationship describes the arrangement; perception records expressed views.

PEOPLE.CSV

CRM | One person, including eligible prospects who never purchase.

Identifiers	Engagement	Relationship	Perception
person_id age_band household_type region	Not observed	record_created_at attributes_recorded_at	primary_need

SUBSCRIPTION_PERIODS.CSV

Renewal system | One scheduled monthly cycle; a paid outcome establishes retained membership for that cycle.

Identifiers	Engagement	Relationship	Perception
period_id subscription_id person_id	Not observed	period_start period_end renewal_due_at renewal_outcome cancellation_effective_at plan_tier	Not observed

TRANSACTIONS.CSV

Finance | One payment or refund event; each payment is one order and refunds reuse its order ID.

Identifiers	Engagement	Relationship	Perception
transaction_id person_id period_id related_transaction_id order_id	occurred_at transaction_type charge_type quantity gross_amount discount_amount refund_amount	unit_price variable_cost	Not observed

PRODUCT_EVENTS.CSV

Product and support | One observed product/support interaction; no event does not mean no subscription.

Identifiers	Engagement	Relationship	Perception
event_id person_id	occurred_at event_type platform	Not observed	Not observed

Each raw field appears once in its dataset's classification. Source-specific rationale is preserved in `diagrams/ucv-tables.json`. A category marked not observed is a data gap, not an invitation to infer a missing attitude or relationship.

Keep campaign credit, response and belief distinct

Technical appendix | Campaign spending and randomized assignment describe different objects from a customer's behavior or survey response.

FEEDBACK.CSV

Customer research | One completed linked survey response; nonresponse is missing, not a zero rating.

Identifiers	Engagement	Relationship	Perception

`response_id`
`person_id`
	`submitted_at`
	`journey_stage`
		Not observed	`ease_rating`
			`value_rating`
			`main_difficulty`

CAMPAIGN_SPEND.CSV

Marketing spending | One campaign per spending date; aggregate before joining to customer outcomes.

Identifiers	Engagement	Relationship	Perception

`campaign_id`
`target_need`
	`spend_date`
	`channel`
	`media_spend`
	`other_acquisition_spend`
		Not observed	Not observed

Campaign-level operational support for acquisition economics, not individual engagement.

ACQUISITION_ATTRIBUTION.CSV

Marketing attribution | One acquired person and the recorded credit for the first successful payment.

Identifiers	Engagement	Relationship	Perception

`person_id`
`first_paid_transaction_id`
`credited_campaign_id`
	`source_type`
	`credited_at`
	`attribution_rule`
		Not observed	Not observed

Measurement metadata for engagement credit, not a customer action itself.

EXPERIMENT_ASSIGNMENTS.CSV

Experiment platform | One eligible person per randomized experiment, including all nonbuyers.

Identifiers	Engagement	Relationship	Perception

`experiment_id`
`person_id`
`tested_campaign_id`
	Not observed	`assigned_at`
		`arm`
		`followup_end_at`
			Not observed

Experimental relationship/design metadata; assignment is not observed exposure or purchase.

SOURCES AND MODEL ADAPTATIONS

Juan Orduz, Revenue Retention and Customer Lifetime Value with NumPyro; Peter Fader and Bruce Hardie, contractual-customer retention and shifted-Beta-Geometric models; Gil Bouhnick, Cohort Analysis. Exact links, adaptation boundaries and licensing are retained in the repository source documentation. Archify is pinned and redistributed with its license notices.

Retention extends Fader-Hardie sBG with mean/concentration parameterization, logit/log covariate links, a non-centered cohort effect on mean churn, grouped conditional Binomial renewal likelihood and NUTS estimation. Revenue adapts Orduz's Gamma structure but uses observed active counts in fitting, an additional per-member shape multiplier, a mean link without an age-by-cohort interaction, need/channel covariates and $100 scaling. Conditional on observed activity, the revenue and retention likelihoods have independent parameter blocks. The separate Orduz-inspired 4-hidden-unit Bayesian neural network is a retention comparison only; its cumulative-count likelihood is changed to conditional renewal trials.

All graphics in this report are generated from this repository's original synthetic records or original diagram specifications. The public sources motivate analytical ideas; their screenshots and private course materials are not redistributed.